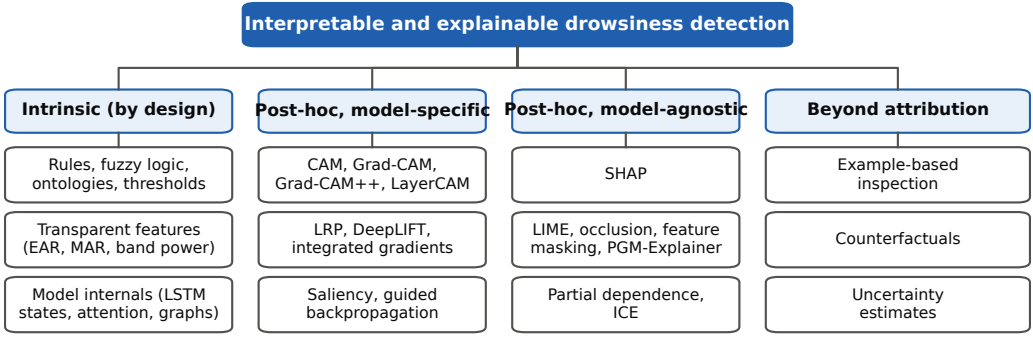
Interpretable and explainable drowsiness detection
Intrinsic (by design)
Post-hoc, model-specific
Post-hoc, model-agnostic
Beyond attribution
Rules, fuzzy logic, ontologies, thresholds
Transparent features (EAR, MAR, band power)
Model internals (LSTM states, attention, graphs)
CAM, Grad-CAM, Grad-CAM++, LayerCAM
LRP, DeepLIFT, integrated gradients
Saliency, guided backpropagation
SHAP
LIME, occlusion, feature masking, PGM-Explainer
Partial dependence, ICE
Example-based inspection
Counterfactuals
Uncertainty estimates
What the explanation refers to
Spatial (channels, regions, connections)
Spectral (frequency bands)
Temporal (time segments)
Face and body (eyes, keypoints)
Features (HRV, blink, steering)
Decision logic (rules)